1.

```
from google.colab import files
import pandas as pd
import numpy as np
import io

# Upload the CSV dataset
uploaded = files.upload()
df = pd.read_csv(io.BytesIO(next(iter(uploaded.values()))))

print("Dataset:")
print(df)

# Select only subject columns
subjects = ["Math", "Science", "English", "History"]
student_scores = df[subjects].to_numpy(dtype=float)

# Calculate average marks
averages = np.mean(student_scores, axis=0)

print("\nAverage marks:")
for subject, avg in zip(subjects, averages):
    print(subject, ":", round(avg, 2))

# Find subject(s) with highest average
highest = np.max(averages)
print("\nHighest average:", highest)
print("Subject(s) with highest average:",
      [subjects[i] for i in np.where(averages == highest)[0]])
```

Output

```
Choose Files  stu (2) - stu (2).csv
stu (2) - stu (2).csv(text/csv) - 753 bytes, last modified: 10/14/2026 - 100% done
Saving stu (2) - stu (2).csv to stu (2) - stu (2).csv
Dataset:
   Reg No     Name  Age  Gender  Math  English  Science  History  Geography
0    S001     Alex   19    Male    85     78.0     92.0     80.0       88.0
1    S002    Brian   20    Male    78      NaN     88.0     85.0       90.0
2    S003    Carla   21  Female    92     90.0     85.0     89.0        NaN
3    S004    David   19    Male    86     85.0     90.0     80.0       95.0
4    S005      Eva   22  Female    88     75.0     95.0     80.0       85.0
5    S006    Frank   20    Male    82     78.0     84.0      NaN       88.0
6    S007    Grace   23  Female    90     80.0     88.0     92.0       87.0
7    S008    Harry   19    Male    85      NaN     87.0     80.0       92.0
8    S009     Iris   21  Female    78     85.0     89.0      NaN       90.0
9    S010     Jack   20    Male    90     88.0     92.0     87.0       80.0
10   S011     Kyle   24    Male    82     84.0     85.0     86.0       93.0
11   S012     Liam   22    Male    85     80.0     88.0     87.0       84.0
12   S013      Mia   23  Female    91     92.0      NaN     89.0       90.0
13   S014     Noah   21    Male    87     86.0     89.0     88.0       85.0
14   S015   Olivia   22  Female    84     87.0     80.0     92.0       88.0
15   S016    Peter   19    Male    88     92.0     85.0     90.0       83.0
16   S017    Quinn   21  Female    90     89.0     88.0     80.0       82.0
17   S018     Ryan   23    Male    92     85.0     90.0     80.0       85.0
18   S019    Sarah   20  Female    87     80.0     85.0     91.0       88.0
19   S020      Tom   22    Male    90     80.0     90.0     85.0       87.0

Average marks:
Math : 86.5
Science : nan
English : nan
History : nan

Highest average: nan
Subject(s) with highest average: []
```

2.

```
from google.colab import files
import pandas as pd
import numpy as np
import io

uploaded = files.upload()
df = pd.read_csv(io.BytesIO(next(iter(uploaded.values()))))

print("Dataset:")
print(df)

sales = df.select_dtypes(include=np.number).to_numpy()
print("\nAverage price:", round(np.mean(sales), 2))
```

Output

```
Choose Files  Sales (1).xlsx - Sales.csv
Sales (1).xlsx - Sales.csv(text/csv) - 104775 bytes, last modified: 10/14/2026 - 100% done
Saving Sales (1).xlsx - Sales.csv to Sales (1).xlsx - Sales.csv
Dataset:
     Product_ID     Month_sales Sales_Rep Region    Total  Quantity_Sold  \
0          1052   February 2023       Bob  North  5053.97             18
1          1093      April 2023       Bob   West  4384.02             17
2          1015  September 2023     David  South  4631.23             30
3          1072     August 2023       Bob  South  2167.94             39
4          1061      March 2023   Charlie   East  3750.20             13
..          ...             ...       ...    ...      ...            ...
995        1010      April 2023   Charlie  North  4733.88              4
996        1067  September 2023       Bob  North  4716.36             37
997        1018      April 2023     David  South  7629.70             17
998        1100   December 2023     David   West  1629.47             39
999        1086     August 2023     Alice   East  4923.93             48

    Product_Name  Unit_Cost    Price Customer_Type  Discount Payment_Method  \
0      Furniture     152.75   267.22     Returning      0.09           Cash
1      Furniture    3816.39  4209.44     Returning      0.11           Cash
2           Food     261.56   371.40     Returning      0.20  Bank Transfer
3       Clothing    4330.03  4467.75           New      0.02    Credit Card
4    Electronics     637.37   692.71           New      0.08    Credit Card
..           ...        ...      ...           ...       ...            ...
995         Food    4943.03  5442.15     Returning      0.29           Cash
996     Clothing    1754.32  1856.40           New      0.21  Bank Transfer
997     Clothing     355.72   438.27     Returning      0.06  Bank Transfer
998  Electronics    3685.03  3743.39           New      0.01  Bank Transfer
999         Food    2632.58  2926.68     Returning      0.14           Cash

    Sales_Channel Region_and_Sales_Rep
0          Online           North-Bob
1          Retail            West-Bob
2          Retail          South-David
3          Retail            South-Bob
4          Online         East-Charlie
..            ...                 ...
995        Online       North-Charlie
996        Retail           North-Bob
997        Online          South-David
998        Online          West-David
999        Online           East-Alice

[1000 rows x 14 columns]
```

3.

```
from google.colab import files
import pandas as pd
import numpy as np
import io

# Upload the sales dataset
uploaded = files.upload()
df = pd.read_csv(io.BytesIO(next(iter(uploaded.values()))))

print("Dataset:")
print(df.head())

# Convert the Total column to numeric
sales = pd.to_numeric(df["Total"], errors="coerce")

# Calculate average sales amount
average_sales = np.mean(sales.dropna().to_numpy())

print("\nAverage sales amount:", round(average_sales, 2))
```

Output

```
Choose Files  Sales (1).xlsx - Sales.csv
Sales (1).xlsx - Sales.csv(text/csv) - 104775 bytes, last modified: 10/14/2026 - 100% done
Saving Sales (1).xlsx - Sales.csv to Sales (1).xlsx - Sales (1).csv
Dataset:
   Product_ID     Month_sales Sales_Rep Region    Total  Quantity_Sold  \
0        1052   February 2023       Bob  North  5053.97             18
1        1093      April 2023       Bob   West  4384.02             17
2        1015  September 2023     David  South  4631.23             30
3        1072     August 2023       Bob  South  2167.94             39
4        1061      March 2023   Charlie   East  3750.20             13

  Product_Name  Unit_Cost    Price Customer_Type  Discount Payment_Method  \
0    Furniture     152.75   267.22     Returning      0.09           Cash
1    Furniture    3816.39  4209.44     Returning      0.11           Cash
2         Food     261.56   371.40     Returning      0.20  Bank Transfer
3     Clothing    4330.03  4467.75           New      0.02    Credit Card
4  Electronics     637.37   692.71           New      0.08    Credit Card

  Sales_Channel Region_and_Sales_Rep
0        Online            North-Bob
1        Retail             West-Bob
2        Retail           South-David
3        Retail             South-Bob
4        Online          East-Charlie

Average sales amount: 5019.27
```

4.

```python
from google.colab import files
import pandas as pd
import numpy as np
import io

# Upload the CSV dataset
uploaded = files.upload()
df = pd.read_csv(io.BytesIO(next(iter(uploaded.values()))))

print("Dataset:")
print(df.head())

# Convert sales amount to numeric
df["Total"] = pd.to_numeric(df["Total"], errors="coerce")

# Extract month from Month_sales
df["Month_sales"] = pd.to_datetime(
    df["Month_sales"], format="%B %Y", errors="coerce"
)

# Calculate total sales for each month
monthly_sales = df.groupby(
    df["Month_sales"].dt.month
)["Total"].sum()

print("\nMonthly sales:")
print(monthly_sales)

# Calculate total annual sales
total_sales = np.sum(df["Total"].dropna())
print("\nTotal annual sales:", round(total_sales, 2))

# Calculate percentage increase from January to December
if 1 in monthly_sales.index and 12 in monthly_sales.index:
    jan_sales = monthly_sales.loc[1]
    dec_sales = monthly_sales.loc[12]

    if jan_sales != 0:
        percentage = ((dec_sales - jan_sales) / jan_sales) * 100
        print("Percentage change from January to December:",
              round(percentage, 2), "%")
    else:
        print("Cannot calculate percentage: January sales are zero.")
else:
    print("January or December sales data is missing.")
```

Output

```
Choose Files  Sales (1).xlsx - Sales.csv
Sales (1).xlsx - Sales.csv(text/csv) - 104775 bytes, last modified: 10/14/2026 - 100% done
Saving Sales (1).xlsx - Sales.csv to Sales (1).xlsx - Sales (2).csv
Dataset:
   Product_ID     Month_sales Sales_Rep Region    Total  Quantity_Sold  \
0        1052   February 2023       Bob  North  5053.97             18
1        1093      April 2023       Bob   West  4384.02             17
2        1015  September 2023     David  South  4631.23             30
3        1072     August 2023       Bob  South  2167.94             39
4        1061      March 2023   Charlie   East  3750.20             13

  Product_Name  Unit_Cost    Price Customer_Type  Discount Payment_Method  \
0    Furniture     152.75   267.22     Returning      0.09           Cash
1    Furniture    3816.39  4209.44     Returning      0.11           Cash
2         Food     261.56   371.40     Returning      0.20  Bank Transfer
3     Clothing    4330.03  4467.75           New      0.02    Credit Card
4  Electronics     637.37   692.71           New      0.08    Credit Card

  Sales_Channel Region_and_Sales_Rep
0        Online            North-Bob
1        Retail             West-Bob
2        Retail          South-David
3        Retail            South-Bob
4        Online         East-Charlie

Monthly sales:
Month_sales
1     495420.37
2     368919.36
3     402638.77
4     438992.61
5     389078.76
6     418458.34
7     374242.88
8     443171.28
9     367837.60
10    460378.78
11    467482.90
12    392643.58
Name: Total, dtype: float64

Total annual sales: 5019265.23
Percentage change from January to December: -20.75 %
```

5.

```python
from google.colab import files
import pandas as pd
import numpy as np
import io

# Upload the fuel efficiency dataset
uploaded = files.upload()
df = pd.read_csv(io.BytesIO(next(iter(uploaded.values()))))

print("Dataset:")
print(df.head())

# Select the highway fuel efficiency column
fuel_efficiency = pd.to_numeric(
    df["highway_mpg(Fuel Efficiency)"],
    errors="coerce"
).dropna().to_numpy()

# Calculate average fuel efficiency
average = np.mean(fuel_efficiency)

print("\nAverage fuel efficiency:",
      round(average, 2), "MPG")

# Compare the first and last car records
model1 = fuel_efficiency[0]
model2 = fuel_efficiency[-1]

if model1 != 0:
    improvement = ((model2 - model1) / model1) * 100
    print("First car efficiency:", model1, "MPG")
    print("Last car efficiency:", model2, "MPG")
    print("Percentage improvement:",
          round(improvement, 2), "%")
else:
    print("Cannot calculate percentage because the first value is zero.")
```

Output

```
Choose Files  Fuel_Efficie...eet1 (1).csv
Fuel_Efficiency datasets.xlsx - Sheet1 (1) - Fuel_Efficiency datasets.xlsx - Sheet1 (1).csv(text/csv) - 38036 bytes, last modified: 10/14/2026 - 100% done
Saving Fuel_Efficiency datasets.xlsx - Sheet1 (1) - Fuel_Efficiency datasets.xlsx - Sheet1 (1).csv to Fuel_Efficiency datasets.xlsx - Sheet1 (1) - Fuel_Efficiency datasets.xlsx - Sheet1 (1).csv
Dataset:
   city_mpg                      class  combination_mpg  cylinders  \
0        25                midsize car               29        4.0
1        26                midsize car               30        4.0
2        25  small sport utility vehicle               27        4.0
3        26  small sport utility vehicle               29        4.0
4        26  small sport utility vehicle               28        4.0

   displacement drive fuel_type  highway_mpg(Fuel Efficiency)   make  \
0           2.5   fwd       gas                            36  mazda
1           2.5   fwd       gas                            37  mazda
2           2.5   fwd       gas                            31  mazda
3           2.0   fwd       gas                            34  mazda
4           2.0   fwd       gas                            32  mazda

       model transmission  year
0          6            m  2014
1          6            a  2014
2  cx-5 2wd            a  2014
3  cx-5 2wd            m  2014
4  cx-5 2wd            a  2014

Average fuel efficiency: 28.61 MPG
First car efficiency: 36 MPG
Last car efficiency: 24 MPG
Percentage improvement: -33.33 %
```